

Bisheng Wang

Email: bishengwangmw@gmail.com | Phone: +351 928276071 | LinkedIn: [in](#)

SUMMARY	I am an Assistant Researcher at INESC TEC in Porto, Portugal, working on computer vision with a focus on medical image analysis. My research experience spans object detection and tracking, medical image segmentation, and visual-language models, developed through academic and industrial experience in China, Austria and Portugal. I am particularly driven to extend these capabilities toward more robust, uncertainty-aware visual systems for complex, high-dimensional data in scientific imaging.	
EDUCATION	Doctor Nanjing University of Science and Technology , China	09/2016 - 06/2022
	Thesis: Pedestrian Detection and Tracking in Crowded Scenes Based on Deep Learning.	
	Bachelor Nanjing University of Science and Technology , China	09/2012 - 06/2016
	Major: Computer Science and Technology. GPA: 3.51/4	
VISITING	Graz University of Technology , Graz, Austria. <i>Visiting PhD Student.</i>	10/2019 - 10/2021
EXPERIENCE	University of Technology Sydney , Sydney, Australia. <i>Visiting Student.</i>	02/2016
	Chang Gung University , Taiwan. <i>Exchange Student.</i>	02/2014 - 07/2014
SKILLS & LANGUAGE	Tools: Python, Pytorch, Numpy, Pandas, Matplotlib, Scikit-learn, OpenCV, Transformers, JuPyter Algorithms: Deep learning, Machine Learning, CNN, RNN, Transformers, LLMs Interests: Medical Image Segmentation, Object Detection and Tracking, Vision Language Model Language: Mandarin (Native), English (Fluent, C1)	
WORKING EXPERIENCE	1. INESC TEC , Porto, Portugal. <i>Assistant Researcher.</i>	03/2025 - Present
	(1) Conducting research on OCTA image segmentation and cell segmentation.	
	(2) Preparing and contributing to national and EU project proposals.	
	(3) Coordinating technical tasks and supervising junior researchers and students.	
	2. Huawei , Nanjing, China. <i>Senior AI Algorithm Engineer.</i>	07/2022 - 07/2024
	(1) Served as module leader for Data Augmentation in Huawei eDataMate, and core member of Data Management in Huawei DME.	
	(2) Delpoyed NLP and computer vision techniques (e.g., OCR, NER, prompt engineering) in Huawei's commercial data platforms. Filed two patents.	
	(3) Participated in project proposal writing and application. Supervised junior engineers and interns.	
	3. The Third Research Institute of Public Security , Shanghai, China. <i>Intern.</i>	11/2017 - 06/2019
	(1) Developed real-time systems for student altercation detection and crowd counting; published 1 peer-reviewed paper.	
PROJECT EXPERIENCE	(2) Gained solid experience in Python, PyTorch, and MLOps for applied AI productization.	
	1. Huawei eDataMate , <i>Technical Leader of Data Augmentation Module.</i>	07/2023 - 05/2024
	Led design and implementation of data tool chain for Huawei's AI data platform. Contributed to project proposal writing and inter-team coordination. System deployed in internal enterprise applications.	
	2. Huawei DME , <i>Core Member - Data Management Module.</i>	10/2022 - 06/2023
	Developed key algorithms for data classification, data quality evaluation, and NER. Co-authored proposal documents for project application. Supervised junior engineers and one patent filed.	
	3. INTERACT (879648), <i>Austrian FFG-funded project.</i>	11/2019 - 06/2021
	Worked closely with Austrian partners on visual recognition pipelines.	
	4. Hyperspectral Image Classification (BK20191284). <i>Provincial Natural Science Foundation Project.</i>	07/2019 - 06/2022
	Focused on weakly-supervised learning and graph-based modeling for spectral data. Contributed to algorithm design and implementation.	
	5. Dongyang Safe Campus "Smart Eye" Project (DYCG2018-A032). <i>Algorithm Developer; Government Project.</i>	09/2017 - 06/2018
	Co-led project proposal writing and technical presentations to local authorities. Designed multi-object tracking and altercation detection algorithms for real-time surveillance.	
	6. Ruijin City Digital Urban Management (1171061611) . <i>Core Technical Contributor.</i>	10/2016 - 05/2017
	Developed AI-based solutions for urban environment monitoring and resource dispatch.	

Google Scholar: Link, citations: 590+, h-index: 9	
PUBLICATIONS	[1] Wang B, Cardoso J S, Wu L. Personalized Cell Segmentation: Benchmark and Framework for Reference-Guided Cell Type Segmentation[C]//ICASSP 2026-2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2026: 7532-7536. [2] Wang B, Cao G, Shang Y, et al. Single-column CNN for crowd counting with pixel-wise attention mechanism[J]. Neural Computing and Applications, 2020, 32: 2897-2908. [3] Wang B, Cao G, Zhou L, et al. Task differentiation: Constructing robust branches for precise object detection[J]. Computer Vision and Image Understanding, 2020, 199: 103030. [4] Wang B, Fruhwirth-Reisinger C, Possegger H, et al. Drt: detection refinement for multiple object tracking[C]//32nd British Machine Vision Conference: BMVC. 2021.. [5] Wang B, Possegger H, Bischof H, et al. Sparse Message Passing Network with Feature Integration for Online Multiple Object Tracking[J]. arxiv preprint arxiv:2212.02992, 2022. [6] Zhang Y, Cao G, Wang B, et al. Dual sparse representation graph-based copropagation for semisupervised hyperspectral image classification[J]. IEEE Transactions on Geoscience and Remote Sensing, 2021, 60: 1-17. [7] Chen M, Wang B, Cao G, et al. Crowd counting method based on pixel-level attention mechanism[J]. Journal of Computer Applications, 2020, 40(1): 56. [8] Zhang Y, Cao G, Wang B, et al. A novel ensemble method for k-nearest neighbor[J]. Pattern Recognition, 2019, 85: 13-25. [9] Cao G, Wang B, Xavier H C, et al. A new difference image creation method based on deep neural networks for change detection in remote-sensing images[J]. International Journal of Remote Sensing, 2017, 38(23): 7161-7175.
PATENTS	[1] Wang B, Wang Y, Zhou Y. Recognition Method, Device, Storage Medium, and Equipment for Vehicle Window Throwing Behavior, 20.09.2024. Patent Number: CN118675224A. [2] Yu B, Wang B, Wu L. Image Coding Method, Device, Computer Equipment, Chip, and Codec System, 03.12.2024. Patent Number: CN119071485A. [3] Wu L, Wang B, Yu B, Jia Z. A rule generation method and related device. 16.09.2025. Patent Number: CN120653979A. [4] Wang B, Wu L. A method, apparatus, and computing device for data classification management. 23.09.2025. Patent Number: CN 120687611A.
HONORS & AWARDS	1. Two-year scholarship by the China Scholarship Council for international study. 10/2019 - 10/2021 2. Multiple First-Class Scholarship from Nanjing University of Science and Technology. 3. Chairman of 'International Communication Association'. 09/2014 - 06/2015
INTERESTS	Trekking, finished the <i>Camino de Santiago</i> (Spain, 779km) in 2021 Chinese Medicine & Health